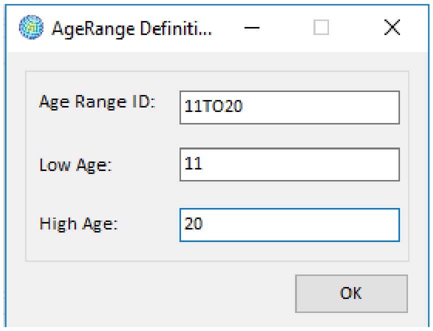
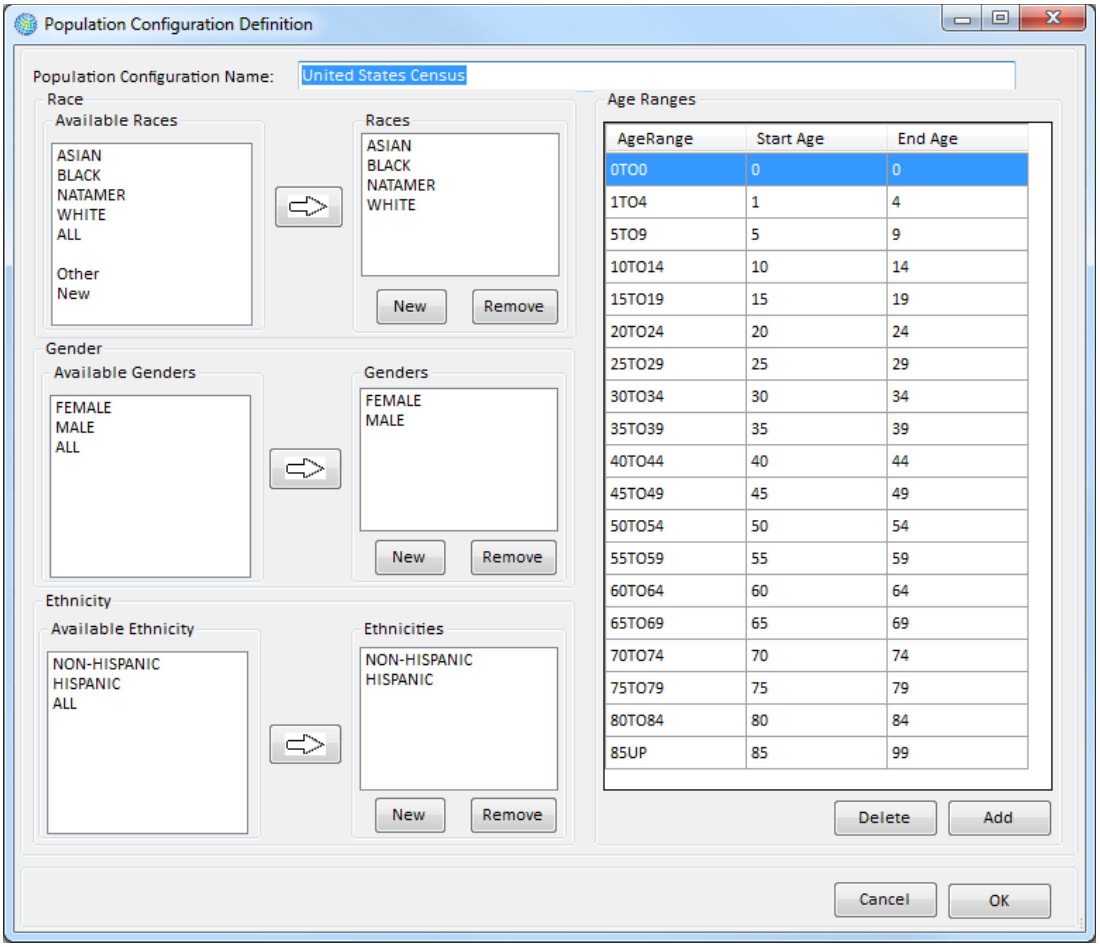
the **Available Races** list box. (If you later create alternative population configurations, you can simply drag the relevant names from the **Available Races** list box into the **Races** list box.) Similarly, under the **Available Genders** and **Available Ethnicity** list boxes, click on **New** and type in the name for any ethnicity and gender identifiers present in your population data.

If you want to remove a selected category from the **Races**, **Genders**, or **Ethnicity** list boxes, then highlight the category that you want to deselect and click the **Remove** button. (It is not possible to delete categories from the **Available Races**, **Available Genders**, or **Available Ethnicity** list boxes.)

The next step is to create the age ranges that match the age ranges in your population file. To start click on the **Add** button below the **Age Ranges** list box. The **AgeRange Definition** window will appear. You do not need totype anything in the **Age Range ID** box. You can enter the upper end of the age range into the **High Age** bound box, which will automatically populate the **Age Range ID**. The **Age Range ID** will be in the format **LowAge**TO**HighAge** (e.g., 0TO0, 1TO17, 18TO25, etc.) For the next age range, BenMAP-CE will automatically fill in the value for the **Low Age** box based on the previous range you entered. For example, the age range names (with corresponding low and high ages) might include the following: *0to0*, *1to4*, *5to9*, *10to14*, *15to19*, *20to24*, *25to29*, *30to34*, *35to39*, *40to44*, *45to49*, *50to54*, *55to59*, *60to64*, *65to69*, *70to74*, *75to79*, *80to84*, and *85up*. You must be sure that the population configuration names exactly match those in your population input file.

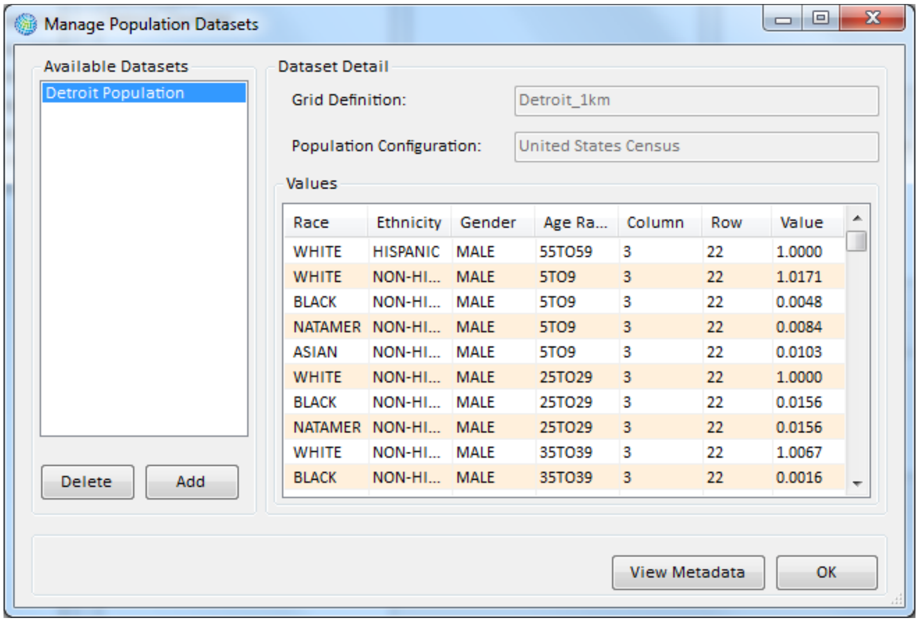


Click **OK** when you have defined the age range. If you make a mistake and want to delete an age definition after you have entered it, select the age range you would like to delete then click on the **Delete** button. The population configurations can be quite detailed, as in the case of the *United States Census* population configuration that comes loaded with BenMAP-CE.



Click **OK** on the **Population Configuration Definition** window to return to the **Load Population Dataset** window.

Click **OK** on the **Load Population Dataset** window to return to the **Manage Population Datasets** window. To view or edit any metadata that was previously added, click the **View Metadata** button. Click **OK** on the **Manage Population Datasets** window. In the **Population Datasets** box of the **Modify Datasets** window you should see an entry for the population dataset that you just loaded.



4.1.5.2 Format for Population Data

Table 4-8 presents the variables that can be used in population datasets. Note that the names you define for age ranges do not need to follow the same pattern used in this manual; the age ranges should be based on what seems most appropriate for you. However, it is critical that the age, race, ethnicity, and gender variables in your population input data exactly match those defined for the population configuration, otherwise BenMAP-CE will fail to load the population data.

Table 4-8. Population Dataset Variables

Variable	Type	Required	Notes
AgeRange	Text	Yes	References a defined age range in the associated Population Configuration.
Column	Integer	Yes	The column and the row link the population data with cells in a Grid Definition.
Row	Integer	Yes	
Year	Integer	Yes	The year of the data. Note that this may include historical population estimates (such as from a census), as well as population forecasts (maximum value = 2499).
Population	Numeric (double)	Yes	Population estimate. Note that the estimate is not restricted to integers.
Race	Text	Yes	References a defined race in the associated Population Configuration.

Variable	Type	Required	Notes
Ethnicity	Text	Yes	References a defined ethnicity in the associated Population Configuration. If no ethnicity is specified in the data, "ALL" should be listed throughout the entire column.
Gender	Text	Yes	References a defined gender in the associated Population Configuration.

4.1.6 Health Impact Functions

Health impact functions calculate the change in the number of adverse health effects among a certain population associated with a change in exposure to air pollution. A typical health impact function has inputs specifying the pollutant; the metric (daily, seasonal, and/or annual); the age, race, ethnicity, and gender of the population affected; and the incidence rate of the adverse health effect.

Fundamental Concept: Health Impact Function or Concentration-Response Function

A **health impact function** calculates the change in the number of adverse health effects associated with a change in exposure to air pollution. The inputs to a health impact function include the change in air quality concentration for a pollutant (using a specified metric such as annual D24HourMean); the size of the affected population (of specified age, race and ethnicity); the baseline incidence rate of the adverse health effect; and an effect coefficient derived from epidemiological studies.

The coefficient for the health impact function is known as Beta (ß) and is derived from epidemiological studies. The value of ß typically represents the percent change in a given adverse health impact per unit of pollution.

Health impact functions are derived from **concentration-response (C-R) functions,** which estimate the relationship between the likelihood of adverse health effects as a function of concentration of an air pollutant. The terms C-R function and health impact function are often used interchangeably.

Health impact functions are subdivided by user-specified types of adverse health effects. The broadest category is the **Endpoint Group**, which represents a broad class of adverse health effects, such as premature mortality, cardiovascular-related hospital admissions, and respiratory- related hospital admissions, among other categories. (BenMAP-CE only allows pooling of adverse health effects to occur within a given endpoint group, as it generally does not make sense to sum or average together the number of cases of disparate health effects, such as premature mortality and chronic bronchitis.) The **Endpoint Group** may then be subdivided by user-specified **Endpoints**. For example, the respiratory-related hospital admission **Endpoint Group**, may have separate **Endpoints** for asthma-related hospital admissions and chronic bronchitis-related hospital admissions.

Fundamental Concept – Endpoint Group

An **endpoint group** represents a broad class of similar or related adverse health effects, such as premature mortality or cardiovascular hospital admissions. An **endpoint** is an individual adverse health outcome or subclass of diagnoses within an endpoint group, identified by one or more codes in the International Classification of Disease system. For example, within the endpoint group *Mortality*, there might be the endpoints *Mortality, Long Term, Lung Cancer* and *Mortality, Long Term, Cardiopulmonary* to distinguish annual mortality impacts with different causes of death.

There are a wide range of variables that can be included in a health impact function, to specify the parameters of the function and to identify its source, such as the **Author**, **Year**, and **Location** of the study, as well as